

NGUYEN MAI HONG TRAM

AI Engineer

📍 Ho Chi Minh City, Vietnam · ✉️ nguyenmaihongtram2004@gmail.com · ☎️ +84 37 585 1104
🌐 github.com/nmhongtram · [linkedin.com/in/nmhongtram](https://www.linkedin.com/in/nmhongtram) · 📁 Portfolio

Profile

Data Science graduate with hands-on experience building and evaluating multimodal AI systems, vision-language pipelines, and production-oriented ML applications.

Education

B.Sc. in Data Science Sep 2022 – Mar 2026
University of Economics Ho Chi Minh City (UEH) GPA: 3.92/4.00

- Rank 2 in Data Science; Top 5% of the School of Business Information Technology.
- Merit Scholarships, 2023 and 2025.
- Selected coursework: Data Structures & Algorithms (9.8/10), Mathematics for Computer Science (9.8/10), Artificial Intelligence (10.0/10).

Selected Programs

Applied AI Talent Program — VinGroup × VinUniversity May 2026 – Aug 2026
Intensive AI training, applied capstone, and industry research internship at VinRobotics
AIO2026 — AI VIETNAM Jun 2026 – Present
Practical artificial intelligence training and community activities

Experience

AI Research Intern Jul 2026 – Aug 2026
VinRobotics — AI Platform, VLA Team Hanoi, Vietnam

- Built and evaluated multi-stage vision-language pipelines for temporally grounded action captioning on egocentric video.
- Designed experiments around frame sampling, temporal grounding, prompting, and model adaptation
- Ran large-scale VLM/LLM experiments and model serving using SGLang, supporting efficient evaluation of multi-modal pipelines.

AI Engineer Intern Mar 2026 – May 2026
TMA Solutions Ho Chi Minh City, Vietnam

- Researched and implemented deep-learning approaches for continuous sign-language recognition and text-to-3D-avatar generation.
- Built an end-to-end inference pipeline from PyTorch models to browser deployment using ONNX Runtime Web.
- Reduced inference time by approximately 37% compared with the previous deployment approach.
- Achieved approximately 5% sequence prediction error on the test set and 10% in a real-time video-chat setting.

Technical Skills

Programming	Python, SQL
AI/ML	PyTorch, Hugging Face Transformers, PyTorch Geometric, scikit-learn, Multimodal AI, VLMs, Computer Vision, RAG
Model Serving & Inference	SGLang, ONNX Runtime, ONNX Runtime Web
Backend & Systems	FastAPI, Celery, Redis, Docker
Data & Retrieval	PostgreSQL, pgvector, Pandas
Language	English — TOEIC Listening & Reading 930/990

Selected Projects

GradeMind — AI-Assisted Mathematics Grading Platform [Demo]

Capstone Project · VinGroup × VinUniversity Applied AI Talent Program

- Built a teacher-in-the-loop AI platform for grading Vietnamese secondary-school mathematics, generating draft scores, evidence-backed feedback, class-level error analysis, and remediation insights.
- Developed a multimodal grading + RAG pipeline with source-grounded retrieval using embeddings and pgvector.
- Designed reliability mechanisms including confidence-based routing, PII safeguards, citation verification, human review, and golden-set evaluation.
- Built the system with FastAPI, PostgreSQL/pgvector, Celery, Redis, React/TypeScript, and Docker.

GNN-SurgVQA — Multimodal Surgical Visual Question Answering [Code]

- Built an end-to-end multimodal VQA pipeline combining visual representations, language models, and graph-based scene reasoning.
- Using SSG-VQA (about 960K QA pairs), achieved 85.9% accuracy and 85.8% weighted F1 on a 77,198-question visual-oracle test.
- Developed an interactive explainability demo with object bounding boxes and attention visualization.
- Work received Best Paper at ICDAM 2026.

Air Quality Forecasting Pipeline [Code]

- Built a forecasting prototype from OpenAQ ingestion and preprocessing to RNN/LSTM/GRU inference.
- Exposed model inference through FastAPI and developed a Streamlit monitoring dashboard.
- Containerized the application using Docker and designed the system for distributed deployment.

Publications & Recognition

GNN-SurgVQA: Object-Centric Graph Reasoning for Visual Question Answering in Laparoscopic Scene Understanding
ICDAM 2026 · Best Paper · [Code]

Comprehensive Approach to Vietnamese Scene Text Recognition: Challenges, Models, and Framework Development
ICDAM 2025 · Springer LNNS, vol. 1601 · [Paper]

UEH Young Researchers Award — Prize B · Eureka 2025 Semifinalist